

[Exp] Multi-Label ICD-10 Classification of Greek Cardiology Discharge Summaries

System Description for the ElCardioCC 2026 Shared Task

Nikolaos Strafiotis, Panteleimon Stanimeros, Vasiliki Katsara, Georgios Chalkias, Glykeria Tsavlidou, Stelios Magalios, Efrosyni Nalmpanti, Panteleimon Stamatakis, and Ioannis Refanidis*

Department of Applied Informatics, University of Macedonia, Thessaloniki, Greece
MSc in Artificial Intelligence & Data Analytics
{aid26012, aid26006, aid26009, aid26004, aid26005, aid26002, aid26014, aid26013}@uom.edu.gr
yrefanid@uom.edu.gr

Abstract. We present our system for the ElCardioCC 2026 shared task on automatic ICD-10 coding of Greek cardiology discharge summaries. The system combines six components: a fine-tuned Greek BERT model, two XLM-RoBERTa variants (base and large), a hybrid information retrieval module, a dictionary-based rule system, and a named entity recognition with entity linking pipeline. These are fused under a metaheuristic ensemble framework. Individual models are trained with task-specific loss functions (Asymmetric Loss, ZLPR) and per-label threshold tuning on the validation set. The ensemble fuses outputs through ten base strategies and four composition operators, with weights searched via random restarts, hill-climbing, and Variable Neighbourhood Search (VNS). Our best submission achieves a micro-F1 of **0.8667** on the official test set, improving by approximately 1.8 F1 points over the best standalone model.

Keywords: ICD-10 coding · clinical NLP · multi-label classification · BERT · ensemble methods · Greek language

1 Introduction

Clinical coding assigns standardised diagnostic codes to free-text medical records, a mandatory step in hospital administration, epidemiological surveillance, and insurance reimbursement. Manual coding is slow, costly, and prone to inter-annotator disagreement, motivating the development of automatic approaches [5].

The **ElCardioCC 2026** shared task targets this problem for Greek cardiology discharge summaries. Systems must predict all applicable ICD-10 codes for each document, a multi-label classification problem over 115 codes.

* Corresponding author.

Three challenges define the setting. First, Greek is a morphologically rich, low-resource language for clinical NLP, with discharge summaries mixing Greek terminology and transliterated abbreviations (PCI, TAVI, CABG). Second, some summaries exceed the 512-token BERT limit. Third, the 115 ICD-10 codes span frequencies from a single instance to over 1,000.

No single architecture dominates across all labels and document types: neural models excel on frequent labels, dictionary rules on procedural codes, retrieval on rare coverage, and NER+EL on mention-level disambiguation. We build a system combining all four paradigms under a **declarative metaheuristic ensemble** that searches over 10 base strategies and 4 composition operators on the validation set.

2 Background

2.1 Transformer Language Models for Clinical NLP

BERT [4] pre-trains a deep bidirectional Transformer [15] using masked language modelling, producing contextual representations that transfer effectively to downstream tasks. Domain-specific variants such as BioBERT [7] consistently outperform general-purpose models on medical benchmarks. For Greek clinical text we use Greek BERT [6] (35K-token vocabulary, pre-trained on Wikipedia, European Parliament proceedings, and web text). XLM-RoBERTa [1] extends RoBERTa [8] to 100 languages with a 250K multilingual SentencePiece vocabulary, naturally handling the Greek–Latin code-switching present in clinical abbreviations (PCI, TAVI, CABG).

2.2 Multi-Label Classification and Loss Functions

Each discharge summary is assigned any subset of 115 ICD-10 codes. A sigmoid-activated linear head produces per-label probabilities; a per-label threshold determines the final binary prediction. Label imbalance is the central challenge: standard BCE is dominated by easy negatives. **Asymmetric Loss (ASL)** [11] applies a stronger focal factor to negatives than to positives, with a probability-shift margin m that suppresses confidently-correct negatives before computing the log. **ZLPR** is a ranking-style objective that directly optimises the margin between positive and negative label logits without requiring calibrated probabilities.

2.3 Retrieval and Ensemble Methods

BM25 and TF-IDF provide sparse retrieval; dense retrieval encodes both query and candidates into a shared embedding space. **RRF** [2] merges ranked lists score-free: $\text{RRF}(d) = \sum_i w_i / (k + \text{rank}_i(d))$, $k = 60$. **VNS** [9] widens the search neighbourhood when no improvement is found, helping avoid local optima. It works well for non-differentiable objectives such as micro-F1.

3 Task and Dataset

3.1 Task Definition

ElCardioCC 2026 is a multi-label ICD-10 coding shared task for Greek cardiology discharge summaries (BioASQ/CLEF 2026), with 115 target codes. The primary metric is **relaxed micro-F1**: predictions from the same clinical cluster as the gold label are not fully penalised, reflecting clinical equivalence among related codes.

3.2 Dataset

The dataset consists of 2,500 anonymised discharge summaries split into training (2,000), validation (250), and test (250) sets, with a separate blind test set held by the organisers. Documents are moderately long: mean 280 tokens ($\approx$ 2,000 characters), maximum 1,218 tokens. Only 2.4% of documents exceed the 512-token BERT limit.

Label distribution. The 115 codes are highly skewed. The five most frequent (Z95, I25, I21, I48, E11) account for $\sim$ 41% of all annotations. At the other extreme, 30 codes appear fewer than 10 times and 4 codes appear only once in the entire dataset (Table 1). This long-tailed distribution severely challenges standard classifiers and requires explicit strategies for rare-label coverage.

Table 1. Label frequency distribution across 115 ICD-10 codes.

Frequency band	# codes	Approx. % of instances
≥ 500	5	41%
100–499	22	36%
10–99	58	21%
< 10	30	$< 2\%$

3.3 Preprocessing and Data Augmentation

Text cleaning. Shared across all splits: whitespace collapse, removal of non-essential special characters (preserving hyphens, commas, slashes, parentheses). Greek BERT additionally receives lowercasing and accent stripping; XLM-RoBERTa receives whitespace-only normalisation to preserve abbreviation signals (“PCI”, “TAVI”). ICD-10 annotations are encoded as 115-dimensional binary multilabel vectors.

Stratified splitting. We apply a custom two-stage Multilabel Stratified Shuffle Split [13] producing an 80/10/10 train/val/test partition. Singleton labels (count=1) are forced to train; labels with 2–4 instances are guaranteed at least one occurrence in both train and test; labels with ≥ 5 instances are guaranteed

at least one in each split. A final patch pass ensures every label seen in val or test is also present in train. Of the 2,500 samples, 1,453 include mention-level annotations enabling NER+EL training.

Synonym augmentation (XLM-RoBERTa base only). Dictionary terms are swapped with clinical synonyms at $p = 0.35$. Multipliers are frequency-adaptive: $\times 4$ (30–99 instances), $\times 3$ (100–199), $\times 2$ (200–299); labels outside this range are left unaugmented.

4 Methodology

The system pipeline is: (i) train six independent components; (ii) tune per-model decision thresholds on the validation set; (iii) search for the best ensemble fusion strategy. Figure 1 provides an overview.

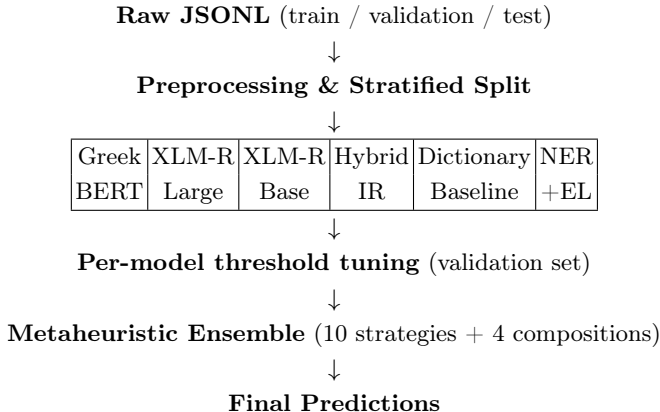


Fig. 1. High-level system pipeline.

4.1 Greek BERT Multi-Label Classifier

Architecture. We fine-tune `nlpaueb/bert-base-greek-uncased-v1` (Greek BERT) [6] with a two-layer MLP head: mean pooling $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(768 $\rightarrow$ 384, GELU) $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(384 $\rightarrow$ 115). Mean pooling outperformed CLS-token extraction by ~ 0.8 F1 points [12], as diagnostic evidence is distributed across the full document. Max sequence length: 384 tokens.

Training. Table 2 lists the final hyperparameters. **ASL** [11] ($\gamma^- = 5$, $\gamma^+ = 1$, clip = 0.05) replaced BCE after Phase 2, raising validation micro-F1 from ≈ 0.74 to 0.788 by suppressing easy negatives. **LLRD** (factor 0.85) updates lower encoder layers at $\text{LR} \times 0.85^l$, preserving pre-trained Greek representations while

adapting upper layers to the clinical domain; gain $\approx +0.4$ F1. The model evolved across four phases (41 WandB runs): BCE baseline (0.74) $\rightarrow$ ASL (0.788) $\rightarrow$ LLRD + MLP (0.811) $\rightarrow$ aggressive tuning (0.827 base, **0.854** after per-class threshold tuning).

Table 2. Greek BERT final hyperparameters.

Hyperparameter	Value
Max length / batch	384 tok / 32 (16 + grad. accum. 2)
Learning rate / LLRD	4×10^{-5} / 0.85 per layer
Warmup / weight decay	6% / 0.02
Loss	ASL ($\gamma^- = 5$, $\gamma^+ = 1$, clip = 0.05)
Epochs / patience	30 / 7 (FP16)

Threshold tuning. A two-pass sweep over $t \in [0.45, 0.75]$ tunes per-label thresholds. Pass 2 accepts per-label values only for labels with ≥ 15 positive examples and $\Delta F1 > 0.001$. This adds $\approx 2\text{--}3\%$ F1, bringing the tuned validation micro-F1 to **0.854** (ensemble context: **0.8165**).

Figures 2–4 show the phase progression, threshold impact, and comparison with XLM-RoBERTa Large.

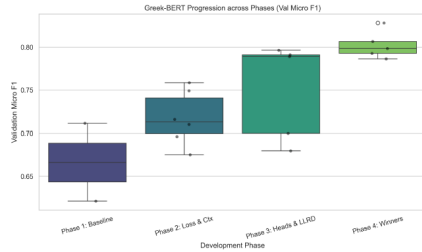


Fig. 2. Validation F1 across finetuning phases.

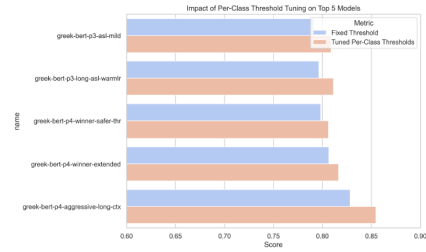


Fig. 3. Per-class threshold tuning impact (+2–3% F1).

4.2 XLM-RoBERTa Classifiers

XLM-RoBERTa’s 250K multilingual vocabulary handles Greek–Latin code-switching better than Greek BERT’s 35K vocabulary.

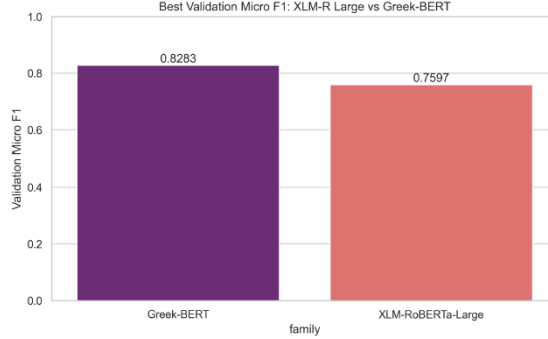


Fig. 4. XLM-RoBERTa Large ceiling (≈ 0.76) vs. final Greek BERT (0.854 tuned).

Long-document strategy. A sliding window (stride 256) handles documents exceeding 512 tokens. At inference, chunk logits are fused: $\hat{y} = 0.5 \bar{y}_{\text{mean}} + 0.5 y_{\text{max}}$.

Large variant. ZLPR loss, effective batch 32 (batch 2, grad. accum. $\times 16$), LR 2×10^{-5} , weight decay 0.05, dropout 0.15, 40 epochs (patience 5). Validation F1 ceiling ≈ 0.76 ; slower convergence and memory constraints (560M parameters) led us to focus development on Greek BERT.

Base variant. Implements *MedicalModelWithDescriptions*: the [CLS] representation passes through 5 parallel dropout layers ($p = 0.1i, i \in \{1 \dots 5\}$) whose logits are averaged (**multi-sample dropout**). A **semantic anchoring** term adds cosine similarity between the document embedding and frozen ICD-10 description embeddings, scaled by a learnable scalar α :

$$\hat{y} = W\mathbf{h} + \alpha \cdot \frac{\mathbf{h} \cdot \mathbf{D}}{\|\mathbf{h}\|} \quad (1)$$

α converges from 0.099 to ≈ 0.12 , confirming a consistent but small semantic correction. Training: AdamW LR 2×10^{-5} , early stopping (patience 4), per-class positive weights (ceiling 20). Post-processing: dynamic threshold tuning + ICD-10 hierarchy enforcement + co-occurrence rules. Validation micro-F1: **0.8101**. Despite using a smaller backbone than the Large variant, the Base model outperforms it (0.810 vs. 0.748) due to the semantic anchoring mechanism and task-specific post-processing, while the Large variant was constrained to batch size 2 under GPU memory pressure.

4.3 Information Retrieval Module

Three channels (BM25 [10], TF-IDF, dense MiniLM embeddings) are combined by RRF [2] ($k = 60$). Codes with RRF score $\geq 0.22 \times \text{top score}$, capped at 12

per document, are predicted. The cap is critical: raising it to 25 drops F1 from 0.662 to 0.140 (Table 3).

Table 3. IR module tuning (validation set).

Configuration	Recall	Precision	F1
Plain top-25	0.374	0.086	0.140
Tuned hybrid	0.692	0.635	0.662

4.4 Dictionary Baseline

A curated CSV (1,680 term-to-code entries) is matched by an Aho–Corasick automaton with a 35-term blacklist. Cardiology rules add codes for named procedures and diagnoses (Table 4); co-occurrence rules enforce clinical consistency (e.g. I22 $\Rightarrow$ I21, I25 $\Rightarrow$ Z95). Validation micro-F1 **0.7176** (precision 0.65, recall 0.80), covering 99.76% of documents.

Table 4. Selected rule-based boosters.

Trigger	Added codes
PCI / PPCI / PTCA / angioplasty	Y84
Stent / pacemaker	Z95
TAVI	Y84, Z95, I35
CABG	Y84, Z95, I25
Myocarditis / pericarditis	I41 / I30
Pulm. embolism / AV block	I26 / I44
ACS / acute coronary syndrome	I20–I22, I24

4.5 Named Entity Recognition and Entity Linking

Greek BERT is fine-tuned as a BIO sequence tagger with a single entity type (MED) and a Partial CRF [14]. The coarse type maximises recall; the Partial CRF enforces label consistency via Viterbi decoding and enables semi-supervised learning from documents with only document-level annotations. Training: 6 epochs, batch 8, LR 2×10^{-5} , early stopping patience 2.

NER spans are augmented by Aho–Corasick dictionary matching; containment-based deduplication retains the longest span. Each mention m is linked to ICD-10 code c by:

$$\text{score}(m, c) = 0.6 \cdot P_{\text{prior}}(c \mid m) + 0.4 \cdot \cos(\mathbf{e}_{\text{ctx}}, \mathbf{e}_c) \quad (2)$$

where P_{prior} is the (mention, code) co-occurrence frequency from training and $\mathbf{e}_{\text{ctx}}$, $\mathbf{e}_c$ are MiniLM embeddings of a ± 200 -char context window and the Greek ICD-10 description respectively. Full document-level dictionary boosting is applied after mention linking. Best validation micro-F1: **0.7223** at epoch 5.

4.6 Metaheuristic Ensemble

Rather than fixing a single fusion rule [3], we define a **declarative strategy space** in YAML and search over it on the validation set, enabling the addition of new strategies without code changes.

Strategy space. The committee consists of the six components above. **Nine base strategies:** (1) *weighted ensemble*: convex combination $\hat{y} = \sum_i w_i \hat{y}_i$; (2) *per-label champion*: route each label to its highest-F1 component; (3) *champion + vote*: champion with majority-vote tiebreaker; (4) *per-patient routing*: highest mean-confidence component per document; (5) *gated secondary*: neural primary, IR/NER fallback for uncertain documents; (6) *frequency-bucketed thresholding*: separate thresholds per label-frequency band; (7) *dual-threshold zones*: high-confidence and borderline zones; (8) *top-K cutoff*; (9) *label correction*: searches (predicted, gold) co-occurrence patterns on the validation set to find label pairs where predicting a should trigger or suppress b ; the resulting rules are applied as post-processing.

Four composed strategies apply label-set algebra over base strategy outputs: *Union* (*merge_or*, maximises recall); *Intersection* (*merge_and*, maximises precision); *k-of-n* (*merge_k2*, predict if $\geq k$ strategies agree).

Weight optimisation. Ensemble weights are searched over the probability simplex $\mathcal{W} = \{w \in \mathbb{R}^6 : \sum_i w_i = 1, w_i \geq 0\}$ directly on validation micro-F1 (non-differentiable) in two stages.

Stage 1 - Random search + hill-climbing: 10,000 weight vectors sampled uniformly from $\mathcal{W}$, each refined by pairwise perturbations accepted if validation F1 improves. Two random restarts help avoid poor local optima.

Stage 2 - VNS [9]: Starting from the hill-climbing solution, VNS runs shake and local-search cycles with increasing neighbourhood radius. When no improvement is found the radius grows; when one is found it resets. This helps find better solutions than pure hill-climbing on a non-smooth objective like micro-F1.

Submitted strategies. The four submissions explore different precision–recall operating points:

1. Per-label routing $\cup$ weighted (union $\rightarrow$ higher recall)
2. Per-label champion+vote $\cup$ weighted (union $\rightarrow$ higher recall)
3. Weighted $\cap$ correction (**best submission**): the label correction module removes false positives from the weighted ensemble output, improving precision.
4. 2-of-3 vote among weighted, per-label routing, and correction.

5 Experimental Results

5.1 Individual Component Performance

Table 5 reports standalone validation micro-F1 for each component. Greek BERT is the strongest single model by a large margin. Dictionary and NER+EL perform comparably; the IR module has the lowest F1 but the highest recall (0.83), making it a key coverage contributor in the ensemble.

Table 5. Individual component validation micro-F1.

Component	Val. micro-F1	Notes
Greek BERT (per-label thresholds)	0.8165	Best single model
XLNet-RoBERTa Base (tuned)	0.8101	Semantic label anchoring
XLNet-RoBERTa Large	≈ 0.76	ZLPR loss; slower convergence
NER + Entity Linking	0.7223	Best at epoch 5
Dictionary Baseline	0.7176	99.76% document coverage
IR (hybrid RRF)	0.6621	Recall 0.69, Precision 0.63

5.2 Ablation Studies

Threshold tuning. Table 6 shows the effect of progressively more granular thresholding. Per-label tuning adds ≈ 1.5 F1 points over a fixed threshold of 0.5 and ≈ 1.0 points over the global optimum.

Table 6. Threshold tuning ablation for Greek BERT.

Strategy	Val. micro-F1
Fixed $t = 0.5$	≈ 0.79
Global sweep $[0.45, 0.75]$	≈ 0.80
Per-label sweep (2-pass)	0.8165

Impact of LLRD. Layer-wise learning rate decay improved validation F1 by ≈ 0.4 points. The improvement is most pronounced for rare labels, where preserving pre-trained lower-layer representations is more important than task-specific adaptation.

Loss function comparison. ASL outperformed both binary cross-entropy and standard focal loss, primarily through better recall on rare labels. BCE delivered good precision but lower recall; focal loss improved recall relative to BCE but less than ASL due to its symmetric treatment of positive and negative examples.

Ensemble composition. Table 7 compares the three composition operators at a schematic level. The intersection (**merge_and**) achieves the best F1 by combining the weighted ensemble’s strong baseline with the correction module’s precision improvement.

Table 7. Ensemble composition operator comparison (validation set, schematic).

Strategy	Recall	Precision	F1 trend
Union (merge_or)	highest	lowest	moderate
Weighted ensemble	mid	mid	strong
k -of- n (merge_k2)	mid	mid+	strong
Intersection (merge_and)	mid–	highest	best

5.3 Official Test Set Results

Table 8 reports all five submissions on the official test set. The best submission uses the **merge_and** composed strategy (weighted ensemble intersected with the label correction module).

Table 8. Official ElCardioCC 2026 test set results.

Submission	Recall	Precision	F1
ensemble (merge_and: weighted + correction)	0.8510	0.8830	0.8667
ensemble (merge_k2: weighted + per-label + corr.)	0.8687	0.8539	0.8613
ensemble (safer thr, merge_k2)	0.8767	0.8431	0.8596
mlc_greek_bert_100 (100% data, no val. split)	0.8194	0.8811	0.8491
mlc_greek_bert (80/10/10 split)	0.8310	0.8675	0.8489

The ensemble outperforms the best standalone model by **+1.8 F1 points** (0.8667 vs. 0.8489), confirming the complementarity of the six components. The full-data Greek BERT (trained on 100% of available data without a validation split) achieves virtually the same F1 as the standard split version (0.8491 vs. 0.8489), suggesting that the validation set contributes little additional signal at this scale.

Among ensemble submissions, the precision-oriented **merge_and** yields the highest overall F1 (0.8667), while **merge_k2** variants achieve higher recall at the cost of precision, consistent with the 2-of-3 voting design. The safer threshold variant of **merge_k2** reaches the highest recall (0.8767) but the lowest precision (0.8431).

5.4 Error Analysis

Rare labels. The 30 codes with <10 training instances are the primary error source. For 10 of these, per-label thresholding is not applicable. The dictionary rule-based system recovers some (e.g. E85 via the amyloidosis rule), but codes with no lexical surface form (e.g. M35, R57) remain consistently unrecovered.

Coverage gaps. 31 label types lie outside the dictionary’s coverage, including metabolic (E00), haematological (D47), and musculoskeletal codes. These require inference from laboratory values or implicit multi-sentence evidence.

Label confusion. I21/I22/I25 (acute MI, subsequent MI, chronic ischaemic disease) frequently co-occur in a single document and require temporal reasoning to assign correctly, which is beyond the capability of current bag-of-token models. Z95 (presence of vascular implants) is the main false-positive source due to ubiquitous stent and pacemaker mentions, motivating the `merge_and` correction step.

Calibration. Greek BERT tends to under-predict rare labels: true positive probabilities for rare codes often fall below 0.5. Per-label thresholding helps but becomes unreliable for labels with fewer than 15 positive validation examples.

6 Conclusion

We presented a multi-component system for automatic ICD-10 coding of Greek cardiology discharge summaries, developed as part of the MSc in Artificial Intelligence and Data Analytics at the University of Macedonia.

The system integrates Greek BERT and XLM-RoBERTa neural classifiers with a dictionary-based rule system, a hybrid information retrieval module, and a NER+EL pipeline. These are fused by a declarative metaheuristic ensemble that searches over ten base strategies and four composition operators using random search, hill-climbing, and Variable Neighbourhood Search. Our best submission achieves micro-F1 **0.8667**, a gain of 1.8 points over the standalone Greek BERT model.

The error analysis identifies three main limitations: rare-label under-coverage due to insufficient training data and calibration issues; dictionary gaps for non-cardiology comorbidity codes; and label confusion among temporally related ICD-10 codes.

Future work will address these through better calibration techniques for the long tail (e.g. temperature scaling, isotonic regression), synonym and back-translation data augmentation to improve rare-label coverage, k-fold cross-validation for more robust model selection, temporal reasoning modules for distinguishing related codes (I21/I22/I25), and cross-lingual transfer from well-annotated English clinical corpora.

References

1. Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., et al. (2020). Unsupervised cross-lingual representation learning at scale. In *Proceedings of ACL 2020* (pp. 8440–8451).
2. Cormack, G. V., Clarke, C. L. A., & Buettcher, S. (2009). Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *Proceedings of SIGIR 2009* (pp. 758–759).
3. Dietterich, T. G. (2000). Ensemble methods in machine learning. In *Multiple Classifier Systems* (pp. 1–15). Springer.
4. Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT 2019* (pp. 4171–4186).
5. Dong, H., Suárez-Paniagua, V., Whiteley, W., & Wu, H. (2022). Automated clinical coding: What, why, and where we are? *npj Digital Medicine*, 5(1), 159.
6. Koutsikakis, J., Chalkidis, I., Malakasiotis, P., & Androutsopoulos, I. (2020). GREEK-BERT: The Greeks visiting Sesame Street. In *Proceedings of the 11th Hellenic Conference on Artificial Intelligence* (pp. 110–117).
7. Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C. H., & Kang, J. (2020). BioBERT: A pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4), 1234–1240.
8. Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., et al. (2019). RoBERTa: A robustly optimized BERT pretraining approach. *arXiv preprint arXiv:1907.11692*.
9. Mladenović, N., & Hansen, P. (1997). Variable neighborhood search. *Computers & Operations Research*, 24(11), 1097–1100.
10. Robertson, S., & Zaragoza, H. (2009). The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4), 333–389.
11. Ridnik, T., Ben-Baruch, E., Zamir, N., Noy, A., Friedman, I., Protter, M., & Zelnik-Manor, L. (2021). Asymmetric loss for multi-label classification. In *Proceedings of ICCV 2021* (pp. 82–91).
12. Mullenbach, J., Wiegrefe, S., Duke, J., Sun, J., & Eisenstein, J. (2018). Explainable prediction of medical codes from clinical text. In *Proceedings of NAACL-HLT 2018* (pp. 1101–1111). ACL.
13. Sechidis, K., Tsoumakas, G., & Vlahavas, I. (2011). On the stratification of multi-label data. In *Proceedings of ECML PKDD 2011* (pp. 145–158).
14. Tsuboi, Y., Kashima, H., Mori, S., Oda, H., & Matsumoto, Y. (2008). Training conditional random fields using incomplete annotations. In *Proceedings of COLING 2008* (pp. 897–904).
15. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., et al. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems 30* (pp. 5998–6008).